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Learning Contact-Rich Assembly Skills Using Residual Admittance Policy*

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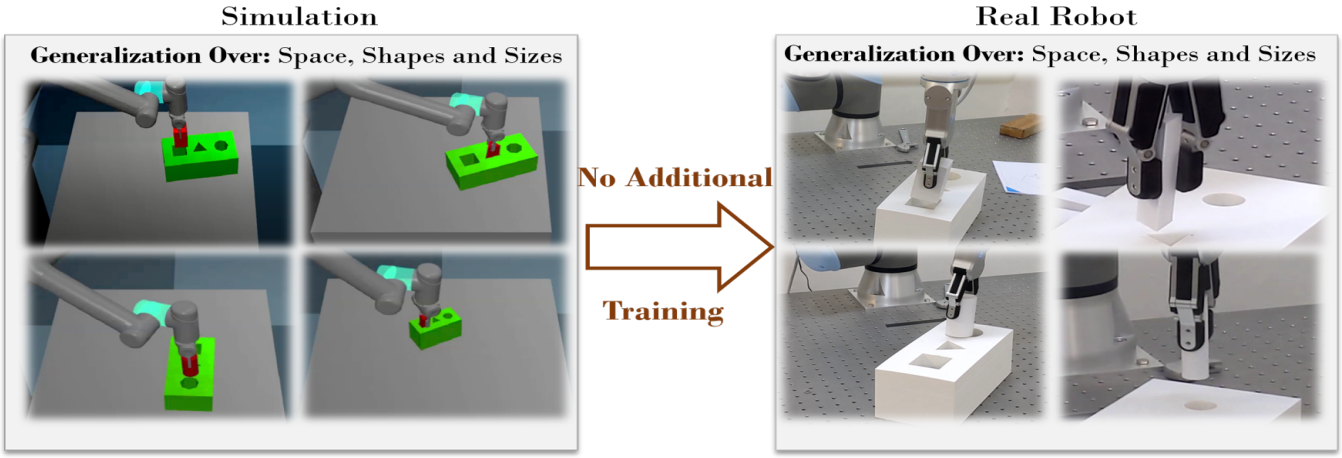


Fig. 1: A residual admittance policy (RAP) trained in simulations to perform peg insertion in different places and with different pegs under uncertainties in board location (left), is deployed successfully on an industrial robot with no further training (UR5e, right). The trained RAP generalizes well over space, shapes and sizes and handles uncertainties in board location in both simulations and real experiments on UR5e.

***Abstract*— Contact-rich assembly tasks may result in large and unpredictable forces and torques when the locations of the contacting parts are uncertain. The ability to correct the trajectory in response to haptic feedback and accomplish the task despite location uncertainties is an important skill. We hypothesize that this skill would facilitate generalization and support direct transfer from simulations to real world. To reduce sample complexity, we propose to learn a residual admittance policy (RAP). RAP is learned to correct the movements generated by a baseline policy in the framework of dynamic movement primitives. Given the reference trajectories generated by the baseline policy, the action space of RAP is limited to the admittance parameters. Using deep reinforcement learning, a deep neural network is trained to map task specifications to proper admittance parameters. We demonstrate that RAP handles uncertainties in board location, generalizes well over space, size and shape, and facilitates quick transfer learning. Most impressively, we demonstrate that the policy learned in simulations achieves similar robustness to uncertainties, generalization and performance when deployed on an industrial robot (UR5e) without further training. See accompanying video for demonstrations.**

I. INTRODUCTION

The role of industrial robots in modern manufacturing has increased significantly in recent decades. However, their limited adaptability and lack of robustness to even small changes [1], hamper further progress. Tasks that require adaptability and robustness, such as assembly and insertion tasks, are mainly performed by human operators. Those tasks are challenging since even small uncertainties in the location of the parts may result in large and unpredictable contact forces. Here we focus on learning haptic-based contact-rich assembly skills for peg insertion that can handle uncertainties and generalize well.

Assembly tasks require position control to achieve the desired relative positions between the assembled parts, and force control to properly respond to contact forces. Simultaneous position and force control in the same axis is impossible [2]. Instead, impedance and admittance control algorithms have been developed to regulate the trade-off between position and force errors [2], [3]. Given the importance of this trade-off for successful assembly tasks [4], [5], the goal of this work is to use RL to learn generalized admittance policy, so the robot can properly perform a wide range of insertion tasks under real-life settings using force and torque feedback.

RL provides promising tools for learning generic robotic manipulxuations [6] and assembly tasks [7]. However, their industrial applications are hampered by sample complexity, which implies that extensive training is needed to learn successful policies that generalize well [8]. Extensive training on physical robots is usually impractical, especially for contact-rich tasks that may cause hazardous forces and wear and tear

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on the robot [8]. One approach is to train in simulations and transfer the trained policy to real world, a phase known as sim2real. However, sim2real is still challenging and usually requires additional training on the physical robot.

In this paper we introduce a method that addresses sample complexity and facilitates generalization, robustness to uncertainties and sim2real, by learning residual admittance policy (RAP). We trained a RAP in simulations to perform peg insertion in different places and with different pegs under uncertainties in board location (Fig. 1). We demonstrate that the trained RAP is robust to localization uncertainties and generalizes well over space, sizes and shapes. Most importantly, we demonstrate that the trained RAP can be deployed successfully on a physical robot (UR5e) with no additional training (Fig. 1, right).

The main contributions of this paper are:

- Learning a task-specific residual admittance policy (RAP) that enables general position and orientation corrections in response to force and torque (F/T) feedback.
- Learning RAP explicitly in terms of general admittance matrices. This facilitates robustness to uncertainties and generalization to new situations that may require different admittance parameters.
- Empirical demonstration that force feedback is sufficient to perform peg insertion given reasonable initial overlap between the peg and the hole.
- Extensive testing demonstrating: (a) Robustness to pose and grasping uncertainties, (b) Generalization over space, size, and shape, (c) Quick transfer learning to new shapes, and (d) Direct transfer of the learned policy to an industrial robot without any further training.

II. TASK DESCRIPTION

Our main goal in this paper is to develop algorithms for learning insertion policies that handle uncertainties in real-world conditions and can generalize well. Thus, we focus on acquiring general relevant skills rather than planning paths for specific problems. In contrast to planners, skills should handle large uncertainties and generalize, at least partially, across: (1) space (hole location), (2) sizes, (3) shapes, and (4) closely related scenarios (e.g., pegs with new shapes).

To facilitate learning, training was performed in simulations using MuJoCo [9] and RoboSuite [10], before testing on an industrial robot, as depicted in Fig. 1. Our primary objective was to learn contact-rich policies that can transfer from simulations to the real world without further training on the physical robot. Hence we separated the processing of vision from F/T feedback. Vision was assumed to provide an erroneous estimate of the pose of the hole, and F/T feedback was used by the residual control policy to correct localization and grasping errors. Thus, in both simulations and real world experiments, the input to the controller was an erroneous pose of the hole, as detailed in Section V-A. The ability of visual systems to localize the hole within the assumed range of errors was verified in Section V-G, but vision was not used otherwise.

Training and testing was performed with pegs of different shapes and sizes and boards at different locations to demonstrate generalization over space, shape and size. Finally, we tested how well the learned policy performs on an industrial robot without further training (Fig. 1).

III. RELATED WORK

A. RL of Insertion Tasks

Despite recent advances in RL methods, their application to robotic tasks is limited due to sample complexity, which requires large amount of experience to learn successful policies [8]. Thus, learning policies that can generalize well over a wide range of conditions is time consuming (hours [11]). Extensive interaction between the robot and the environment due to training is undesired for contact-rich insertion tasks that may cause hazardous forces and result in wear and tear on the robot [8].

The required interaction with the environment can be reduced by more sample efficient methods such as model-based RL or by exploiting prior knowledge about the task using residual learning [8], [12] or learning from demonstration (LfD) [13], [14]. Model-based RL was shown to facilitate one-shot learning, even for contact-rich tasks [15]. However, acquiring a prior model to facilitate generalization requires extensive training [15] while lack of fidelity of contact modeling hampers the integration of haptic feedback [11].

Residual leaning relies on a fixed, well-known baseline policy and learns only a residual correction policy [16]. [8] showed empirically that residual learning outperformed pure RL, especially under perturbations. Focusing on peg insertion, [12] noted that learning the residual policy in every state may result in large actions away from the hole opening and thus suggested to restrict learning to regions close to the hole. Here we learn a residual admittance policy that is active only when interaction forces occur, so it is not affected by, nor does it affect, movements away from the hole.

B. Action Space

The action space used in robotic applications usually involves the desired reference trajectory [8] or desired forces [17]. However, given the importance of force control for contact-rich tasks, it has been suggested to also include the control parameters explicitly, either as the PID gains for hybrid force-position control [18], impedance [19], [20], [5], or admittance parameters [18]. Impedance and admittance are just the inverse of each other and reduce to stiffness and compliance in the static case [3].

In all those cases, the relevant control parameters were learned in addition to the reference trajectory. Furthermore, only single-axis control parameters were learned, i.e., control gains along each of the six axes of the end-effector (rotation and translation), or for individual joints. Focusing on manipulation tasks, [20] suggested to learn a policy that both predicts the reference trajectory and adapts the impedance parameters. Interestingly, learning the impedance gains in end-effector (EEF) space improved sample efficiency and the ability to transfer the learned policy from simulations

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to a real-robot. [5] proposed learning both the impedance and desired position in joint space, to provide flexibility in determining the null-space. They learned single-leg hopping and a sliding-contact manipulation task in simulations, and showed direct transfer of hopping to a real system.

Here we propose to learn only the control gains and impedance parameters to generate a residual admittance policy (RAP), while relying on a baseline policy to generate the nominal trajectory. Thus, RAP corrects the trajectories generated by the baseline policy, but only when actual interaction forces are sensed. Furthermore, RAP is based on generalized admittance, with full rather than diagonal matrices, so it can make general movement corrections. To facilitate generalization, the impedance parameters are learned in EEF space.

C. Dynamic Movement Primitives

We integrate the residual admittance policy and the baseline policy within the framework of Dynamic Movement Primitives (DMP). DMP generates desired trajectories as solutions of non-linear dynamical systems [21]. The linear part of the dynamical system guarantees stability, while the non-linear terms modulate the behavior as desired without affecting stability. The non-linear parts can be learned to achieve the desired behavior, usually via LfD.

The framework of DMP facilitates the integration of different movement primitives [22], which can be used to divide the task into known and unknown control policies for dimensional reduction. The DMP framework has been extended to perform feed-forward or feedback force control using LfD [23], [24]. Specifically, [23] learned to predict the force measurements for feed-forward control, with fixed compliance parameters along individual axes. [24] added a feedback compliance term that responds to deviations between the actual and desired force with manually adjusted gains.

IV. CONTROL METHOD

The proposed control architecture, depicted in Fig. 2, involves both a baseline policy, which generates minimum-jerk trajectories, and a residual admittance policy (RAP), which is learned using deep RL to generate trajectory corrections in response to sensed F/T. The F/T that occur when the EEF contacts the hole's edge are related to the deviation from the hole. RAP captures this relationship for a given hole's shape and location to generate trajectory corrections. RAP can be learned in simulations to avoid hazardous contact forces on the robot and to facilitate generalization over different environments. Furthermore, we demonstrate that learning RAP facilitates direct transfer from simulations to the physical robot without any further training.

A. Baseline Policy

Using the framework of DMP, the baseline policy is generated by adding a proper non-linear forcing term f to a linear second-order dynamical system, which has a single attractor at the goal g [21]:

$$\tau\ddot{y} = K_y(g - y) - D_y\dot{y} + (g - y_0)f \tag{1}$$

where K_y and D_y are spring and damping diagonal matrices, respectively, defining the PD gains, and τ is a scaling factor. In our case, y is a six-dimensional vector $y = [x \quad \theta]'$ that includes the 3-dimensional position x and 3-dimensional orientation θ of the end-effector (EEF) in Cartesian space.

The non-linear function f can be expressed as a weighted sum of Gaussian basis functions [21]. The weights can be learned to generate desired trajectories, using locally weighted regression. Thus, f can be designed to manage the known part of the task, and, in particular, is designed here to generate minimum jerk trajectories [25], [26]. The residual admittance policy, considered next, is designed to modify those trajectories in response to interaction forces.

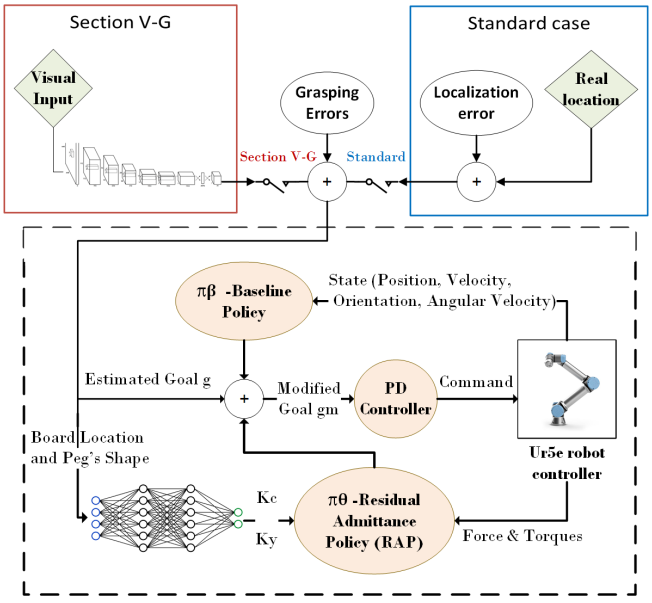


Fig. 2: Control architecture integrating a baseline policy π_β and a residual admittance policy (RAP) π_θ, to modify the goal g set by the task planner. The input to the integrated controller is the estimated location of the board that is derived from either: (1) the real location plus localization error (Section V-A, standard case), or (2) simulated vision system (Section V-G). RAP is defined by the control gains K_y and compliance parameters K_c that are determined by a NN given the estimated board location and peg type. The modified goal is followed by a PD controller that sends commands to the robot.

B. Residual Admittance Policy

Admittance control modulates the trajectory to achieve a proper balance between deviations from the reference trajectory and interaction forces [2], [3]. In particular, the EEF can be controlled to behave as a desired spring and damper system in Cartesian space:

$$\Delta F = K_c\Delta y + D_c\Delta\dot{y} \tag{2}$$

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where $F \in \mathbb{R}^6$ is the F/T vector in EEF space, ΔF is the deviation from the desired F/T, and Δy is the deviation from the reference trajectory. The 6×6 matrices K_c and D_c are the generalized stiffness and damping matrices. We do not restrict K_c and D_c to be diagonal nor block-diagonal, so they can provide generalized relationships. In particular, off-diagonal terms can couple deviations in one direction with F/T measurements in another direction as well as couple angular deviations with forces, or transitional deviations with torques. Here we assume that the desired F/T is zero, so $F = \Delta F$.

Given the small velocities expected during insertion tasks, we focus on stiffness control with $D_c = 0$, so Eq. (2) can be inverted directly to compute the deviations:

$$\Delta y = K_c^{-1} F \tag{3}$$

The original dynamical system, Eq. (1), is now re-written in terms of the modified trajectory, $g + \Delta y$, rather than the reference trajectory, g:

$$\tau \ddot{y} = K_y(g - y + \Delta y) - D_y \dot{y} + (g - y_0) f_\omega \tag{4}$$

where f_ω is the nonlinear term that generates minimum-jerk trajectories.

Given Eq. (3), Eq. (4) has the form of Eq. (1) with a novel compliance coupling term c_c:

$$\tau \ddot{y} = K_y(g - y) - D_y \dot{y} + (g - y_0) f_\omega + \boldsymbol{c_c} \tag{5}$$

The coupling term provides compliance according to:

$$c_c = K_y \Delta y = K_y K_c^{-1} F \tag{6}$$

C. Learning Residual Admittance Policy

The diagonal spring matrix, K_y, and the full stiffness matrix, K_c, were learned using proximal policy optimization (PPO) [27]. We chose a model-free on-policy algorithm since it is relatively less brittle. As mentioned above, we improved sample efficiency by learning a residual admittance policy, which simplifies training and facilitates the transition from simulations to real-world.

The diagonal spring matrix, K_y, was learned since it affects not only the accuracy of position tracking, but also the stiffness during interaction with the environment, especially when hitting the surface. K_y was restricted to be positive definite with the same constants on the upper and lower parts, so $K_y = diag[k_t, k_t, k_t, k_\theta, k_\theta, k_\theta]$, where $k_t, k_\theta > 0$ are the translation and rotation gains. The damping matrix D_y was determined from the spring matrix to achieve critical damping for both translation and rotation.

The key element of RAP is the full 6×6 stiffness matrix, K_c. K_c was not restricted to be symmetric, so in addition to the conservative spring-like force-field generated by symmetric stiffness matrices, it may generate also and a non-conservative rotational force-field [28]. Furthermore, K_c was not restricted to be positive semi-definite since the direction of movement correction relative to the F/T may depend on contact configuration. Thus RL can find the best parameters for the task. While non-positive stiffness would cause increasingly larger F/T in some directions, we expect the rotational part of the stiffness matrix to be able to re-orient the peg to enter the hole before this causes a problem. This issue and the ability and merit of restricting the stiffness to be positive definite will be investigated in the future. Thus, a NN was trained to determine 38 parameters including the gains, k_t and k_θ, and the 36 parameters of the full stiffness matrix, K_c, given hole location, orientation and shape (Fig. 2).

D. Reward

Reward shaping is a delicate craft, especially for general skills. We divided the reward function to 2 parts: (1) Outcome reward, of a fixed 38000 points, given at the end of successful trials, and (2) Shaped reward given for unsuccessful trials. A trial was considered successful if the tip of the peg passed through the hole and reached the table within $\epsilon = 20$[mm] in the allocated time of 60[s]. The shaped reward was a function of the distance between the tip of the peg and the desired destination, and was included to facilitate convergence.

E. Control Implementation

Eq. (5) can be viewed as a PD controller in Cartesian space with a modified goal set by the external nonlinear forcing term f_ω and coupling term c_c, as depicted in Fig. 2:

$$\tau \ddot{y} = K_y(g_m - y) - D_y \dot{y} \tag{7}$$

where, using Eq. (6), the modified goal is given by:

$$g_m = g + K_y^{-1}(g - y_0) f_\omega + \Delta y \tag{8}$$

Following Eq. (7), we force the robot to move at the required EEF acceleration. In simulations, this is accomplished by converting the desired EEF acceleration to joint torques, using standard inverse dynamics method [29]. In robotic experiments, this is accomplished by sending EEF effort commands to the robot.

V. SIMULATION DESIGN AND RESULTS

Simulations were designed to address the following questions: (1) How well can the learned policy handle uncertainties in hole location? (2) Can a single policy learn to perform insertion with pegs of different shapes? (3) How well does the learned policy generalize over peg size? (4) How well does the learned policy generalize to pegs with new shapes? (5) How fast and how well can we achieve transfer learning?

Simulations were conducted in MuJoCo, a Multi-Joint with Contacts physics engine [9], using a UR5e model, by Universal Robots. The force sensor was modeled to mimic the OnRobots HEX-E F/T sensor used in the real robot.

A. Environmental conditions

Training and testing was conducted under different environmental conditions, with the following **physical and Geometrical parameters:**

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- Position: board location was uniformly distributed in the (x, y) plane within the workspace (avoiding singularities). Orientation was uniformly distributed between $0 - 90°$.
- Shape: the cross section of the peg and corresponding hole was a circle (with diameter D), square or isosceles triangle (with edges of length L), so the policy was not trained for a specific shape.
- Size: the characteristic size of the pegs $S = L = D$ was $S = 50$[mm] during training and $S = 25, 50, 60$[mm] during testing. The characteristic size of the holes was $1.03D$ for circular holes and $1.08L$ for other holes. The depth of the holes was $d = 80$[mm].
- Collision and friction: collision between pegs and board was modeled as rigid body interaction in MuJoCo [9] with uniformly distributed friction coefficients between $0.2 - 0.9$, corresponding to steel/steel friction.

In order to handle large uncertainties in localization and grasping, we inserted the following errors:

- Translational errors were distributed uniformly within $\pm$ the indicated range, specified as percent of half the characteristic size (up to 32% of $S/2$).
- Orientation errors were uniformly distributed within $\pm$ the indicated range (up to $12°$).

Note that visual localization uncertainties would result in translational errors in (x, y) plane and orientation errors around z-axis, while grasping uncertainties would result in the other errors. The visual localization assumptions were validated by training a YOLO-lite model based on a built-in camera in the simulation, as detailed in Section V-G.

B. Learning the Nominal Policy

Initial training was limited to cylindrical and cubical pegs, and was conducted with the parameters and errors specified in Section V-A. The training curve depicted in Fig. 4 presents trial-to-trial reward in light blue and average reward, over 100 steps (trials), in bold. Improvement was slow in the initial 6000 steps, but accelerated very quickly over the next 2000 steps. Trial-to-trial return fluctuated due to trial-to-trial changes in linear and angular errors, which were selected randomly for each trial. After reaching above 35000 points, the average reward remained above 33000 and reached above 34000 (92% of the maximum of 38000) at the end of training. The final policy is referred to as the nominal policy and its performance is evaluated below.

C. Nominal Performance

The performance of the nominal policy was evaluated in $N_{test} = 200$ simulated trials in each of over 350 ranges of linear and angular errors. Success-rate and mean time-to-completion in each range are detailed in Fig. 3(a) and Fig. 3(b), respectively. Each trial was initiated with randomly selected initial arm configuration and board location, with uniformly distributed errors within $\pm$ the indicated error-range.

Changes in performance from one error-range to the next are not consistent due to random variations in selected errors and initial conditions. Nevertheless, Fig. 3(a) and Fig. 3(b) suggest that success-rate tends to degrade, while mean time-to-completion tends to increase, with the error size.

Further evaluation and comparison is facilitated by dividing the error-plane into 4 quarters as marked by black lines in Fig. 3. The statistics of success-rate and time-to-completion in each quarter are summarized in Fig. 5. Hypothesis testing was performed for each pair of quarters, evaluating the null hypothesis that the success rates are from the same distribution, against the alternative hypothesis that the mean success rates decrease from the first to the fourth quarters. Hypothesis testing was conducted using one-sided MannWhitney U test with significance level $\alpha = 0.05$. The results indicate that success-rates degraded significantly from the first to the fourth quarter, while mean time-to-completion increased significantly from the first to fourth quarter. Accounting for the effect of the clearance, the results still indicate that the performance improves with localization and grasping accuracy.

D. Generalization Over Different Sizes

As mentioned before the nominal policy was trained on cylindrical and cubical pegs with characteristic size $S = 50$[mm]. Generalization to larger ($S = 60$[mm]) and smaller ($S = 25$[mm]) pegs was evaluated with $N_{test} = 50$ trials at each of 100 error-ranges, and the results are summarized in Fig. 5. Hypotheses testing demonstrates good generalization: Comparing success-rates for large versus nominal pegs, there were no significant differences in the first and third quarters, and thought success-rates were statistically lower in the second and fourth quarters, differences in mean success-rates were less than 2.5% and 4.5%, respectively. Generalization to smaller pegs and holes is more challenging since tracking errors are relatively large. Success rates were statistically lower for small compared to nominal pegs in the first and second quarters, but the differences were less than 3%.

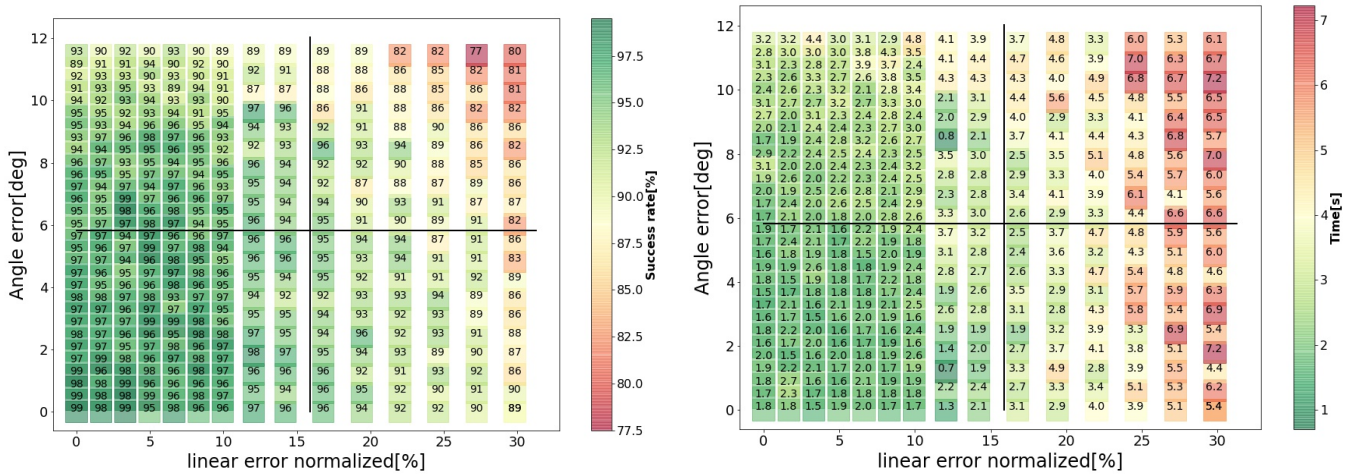
E. Generalization Over Closely-related Scenarios

Generalization over closely-related scenarios was evaluated by testing the performance of the nominal policy on pegs with triangular cross-sections, without any further training. The NN input that specifies the shape was set to cubical. Success rates were evaluated on $N_{test} = 50$ trials at each of 100 error-ranges, and summarized in Fig. 5. While degradation in performance was significant, success rates in the first quarter of the error-plane remained above 66% with average of 75%. Hence, the nominal policy was not over-fitted to cubical pegs, and may provide a good initial policy for learning to insert other pegs, as evaluated next.

F. Transfer Learning

We evaluated transfer learning by testing how fast and how well the nominal policy can be re-trained to insert pegs with another shape. Starting with the nominal policy, Fig. 6 depicts the average return during subsequent training using pegs with triangular cross-sections. It is evident that only 800 trials were required to re-optimize the nominal

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(a) Success rate

(b) Time of completion

Fig. 3: (a) Success rates achieved by the nominal policy during testing as a function of the range of linear and angular errors under the same experimental conditions as those used for training. Success rates in each error-range were estimated from $N_{test} = 200$ trials with randomly selected arm configurations, board locations and uniformly distributed errors within $\pm$ the indicated linear and angular error-ranges. Black lines divide the error-plane into 4 quarters for further evaluation: Q1 (bottom left), Q2 (upper left), Q3 (bottom right) and Q4 (upper right). (b) Mean time-to-completion with the nominal policy.

policy for pegs with triangular cross-sections and improve the average return from around 24000 to around 34000. This is a reduction of an order of magnitude in the number of trials, compared to initial training (Fig. 4).

Re-training resulted in significant improvement in performance as summarized in Fig. 5. Success rates after re-training were statistically higher ($\alpha = 0.05$) than those before re-training in all quarters of the error-plane. Furthermore, success rates in the first quarter of the error-plane were not statistically distinguished from nominal success rates. While success rates in the other quarters of the error-plane were significantly lower than nominal success rates, mean differences were less than $7\%, 6\%$ and 11% in the second, third and fourth quarters, respectively. The lower performance may be attributed to the complexity of inserting pegs with triangular cross-sections.

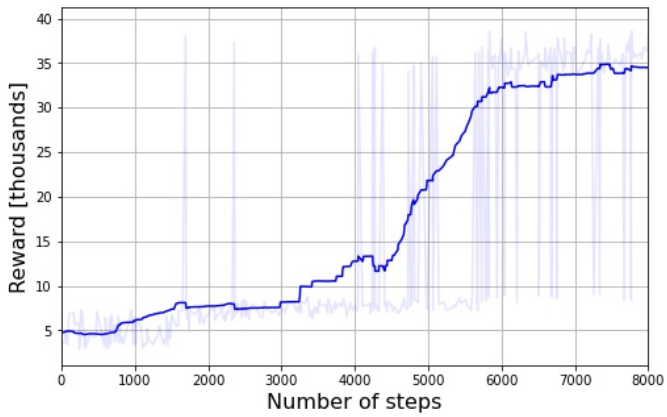


Fig. 4: Training curve: moving average reward over 50 steps (bold) and trial-to-trial reward (light) as a function of steps.

G. Vision to Action

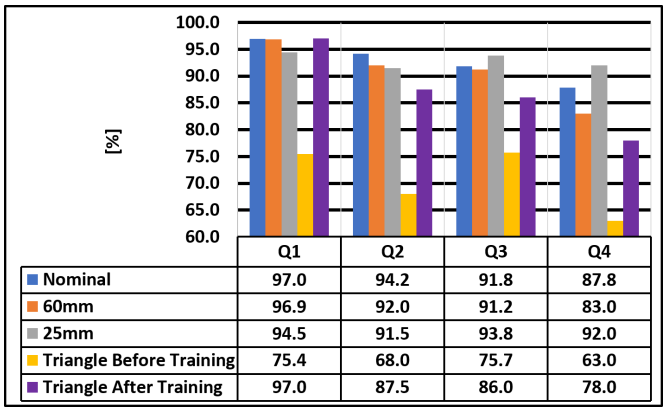
To validate that the range of imposed errors, detailed in Section V-A, is comparable with visual localization errors, we evaluated the nominal policy using object localization derived from synthetic images. Bird's view images, which are useful for planning consecutive actions, were generated by a simple 2D 1Mpx ($1024 \times 1024 \times 3$) simulated camera. A YOLO-lite model [30], detailed in [31], was trained to localize the holes in the board within a maximum error of 8[mm] in x and y and 8° in planar rotation. Grasping uncertainties were accounted for by setting the orientation errors around x and y and location errors along the z-axis to their maximum levels.

Combining the nominal policy with visual object localization resulted in 92% success rate over $N_{test} = 200$ trials with both cylindrical and cubical pegs ($S = 50$[mm]). This result fits well with the nominal success-rate in the third quarter (91.8%, Fig. 5). Better results can be achieved by improving the visual localization algorithm.

VI. SIMULATION-TO-REAL TRANSFER

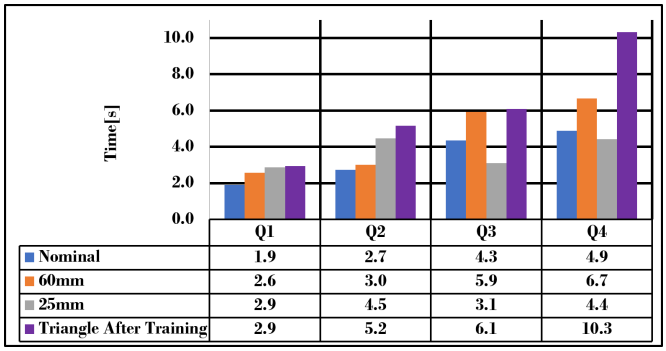
Final experiments evaluated how well the policy that was learned in simulation can be transferred to a physical robot without further training. Experiments were conducted on an industrial robot, UR5e, equipped with OnRobots HEX-E F/T sensor, which was modeled in simulations. To accurately capture the contact forces and generate smooth movements, high frequency communication was used (Real Time Data Exchange). Thus, F/T and position measurements were sampled at 500Hz and effort commands were sent at 125Hz.

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	Q1	Q2	Q3	Q4
Nominal	97.0	94.2	91.8	87.8
60mm	96.9	92.0	91.2	83.0
25mm	94.5	91.5	93.8	92.0
Triangle Before Training	75.4	68.0	75.7	63.0
Triangle After Training	97.0	87.5	86.0	78.0

(a) Success rate (%)



	Q1	Q2	Q3	Q4
Nominal	1.9	2.7	4.3	4.9
60mm	2.6	3.0	5.9	6.7
25mm	2.9	4.5	3.1	4.4
Triangle After Training	2.9	5.2	6.1	10.3

(b) Time of completion (sec)

Fig. 5: Summary of simulation results in terms of success-rate (a) and time-to-completion (in sec) (b), in different conditions and ranges of errors. Q1-Q4 refer to the four quarters of the error-plane as marked in Fig. 3. Triangle before/ after training refer to pegs with triangle cross-section and either the nominal or re-trained policy, respectively, as detailed in Section V-F.

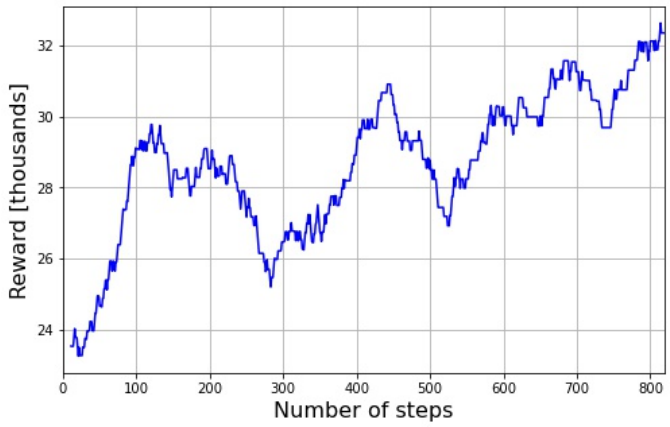


Fig. 6: Re-training curve: moving average reward over 50 steps, during re-training of the nominal policy on pegs with triangular cross-section not used during original training.

The robot inserted six pegs of different shapes and sizes into matching holes in randomly placed boards (as detailed in Section V-A, see right side of Fig. 1 and the supplementary video). The pegs and board were made in 3D printing. Robustness to uncertainties was evaluated by adding stochastic errors to the robot's goal in (x, y) plane and around the z-axis, as described in Section V-A. Furthermore, no effort was made to grasp the peg in a precise height or orientation or to avoid slippage during insertion. The control parameters were determined by the NN trained in simulations, given the erroneous board position and peg type.

TABLE I: Experimental results with UR5e robot: Number of successful trials out of 10 trials at different board locations, for 6 different pegs and 5 ranges of linear errors. Angular errors were uniformly distributed in the range of $\pm 12°$. Empty cells reflect combinations that were not tested.

Peg	**Linear error-range**				
Cross-section (size)	3mm	4mm	6mm	8mm	10mm
Cylindrical (50mm)			10/10	10/10	8/10
Cubical (50mm)			10/10	9/10	9/10
Triangular (50mm)			10/10	7/10	5/10
Cylindrical (25mm)	10/10	10/10	10/10		
Cubical (25mm)	10/10	10/10	9/10		
Triangular (25mm)	10/10	8/10			

Table I summaries the experimental results with UR5e for six different pegs (rows), under different ranges of linear errors (columns). Angular errors were uniformly distributed in the range of $\pm 12°$ in all cases. Each experimental result is based on 10 trials with different board locations and errors. Pegs with the nominal size were tested with large errors while pegs with smaller size were tested only with small errors. The results indicate that the policy learned using the proposed method can be deployed directly on the physical robot with no further training. This can be attributed to a number of unique features of the proposed method, as discussed next.

VII. DISCUSSION

The proposed method is unique in its ability to generalize over space, shape, and size, under a wide range of uncertainties. Furthermore, to the best of our knowledge, our method is the first haptic-based policy to enable direct transfer of the learned policy from simulations to real-life without any further training.

Comparison to classical methods: Recent classical methods, including spiral search combined with contact identification [32], and Gaussian Mixture Model (GMM) derived from human demonstration [33], were demonstrated to achieve high success rate on cylindrical pegs, but, as expected, did not demonstrate generalization over space, size, and shape.

Comparison to model-free RL methods: Learning to assemble blocks with residual RL, [16] considered variations in the alignment of the two side-blocks, but did not address the issue of generalization. They reported quick transfer of the policy initialized in simulations to the real world, but still required additional training on the robot. Learning peg insertion, [12] restricted residual learning to the vicinity of the hole. Learning was performed directly on the physical robot, requiring hours of training. They addressed robustness to uncertainties but not generalization.

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[11] addressed sample complexity and generalization by learning multimodal sensor representation. They demonstrated generalization over shape, but not over space and size. Most importantly, the policy required learning on the robot itself for about 5 hours, which can hamper real life implementation.

Simulation-to-real transfer: The seamless transfer of the learned policy from simulations to real-life may be attributed to five key features of the proposed method: (1) Learning the relevant admittance, which is inherently robust to uncertainties, including those associated with the transition to real-world. (2) Formulating admittance control in Cartesian space [20]. (3) Focusing on learning the residual admittance policy, while relying on a baseline policy in free-space. (4) Adding domain randomization, and in particular, randomly distributed friction coefficients, during training in simulations [34]. (5) The learned policy is affected only by F/T measurements while vision provides only rough object localization.

VIII. CONCLUSIONS

We introduced a novel method for learning insertion tasks for industrial applications, without acquiring data on the real-robot. The proposed method, dubed residual admittance policy (RAP), was demonstrated to handle a wide range of localization and grasping uncertainties and to generalize well over space, size and shape. Most impressively, we demonstrated that the policy learned in simulations was successful in controlling an industrial robot (UR5e) to insert pegs of different shapes and sizes in random locations despite localization and grasping errors, without further training.

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